

ActInf Textbook Group ~ Cohort 2 ~ Meeting 16 (Chapter 7, part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_2 | Cohort 2 ~ Meeting 16 (Chapter 7, part 1) | 55:55

Hello, it's February 15th, 2023. We're in Cohort 2 of the Active Textbook.

We are in our first discussion on Chapter 7.

So, let's just first look at how Chapter 7 is anticipated in the text.

Then, any of you are welcome to give some thoughts or just raise any thoughts or questions on Chapter 7.

And then that will go wherever it goes.

We can also look at previously submitted questions as well as walk through the text and get a big picture of what's happening.

So, the first mention of Chapter 7 is in Chapter 1 when they're summarizing Chapter 7.

And they describe that in Chapter 7, they're going to discuss active inference models addressing problems formulated in discrete times as hidden Markov models and partially observable Markov decision processes.

And they mention the examples they're going to use, which is a left or right at a T maze.

And they also bring in topics, information seeking, learning, novelty seeking.

They talk about how preference represented by the C parameter is going to be unpacked in Chapter 7.

In Chapter 4, they talk about how message passing will be unpacked further in Chapter 7.

In Chapter 5, they talk about E and policy.

So, as they're mentioning a bunch of the parameters E , π , C , message passing, they're describing that they're going to come back to it in Chapter 7.

If we want to change the way the signal is interpreted, we need to rely on learning.

We will return to this in detail in Chapter 7.

In Chapter 7, we will discuss a biologically plausible example of factorization, specifically the what and the where streams.

And in Chapter 6, in the recipe,

they talk about the updating and the form of the updating.

So, several more mentions of Chapter 7 in Chapter 6,

and then we actually get to Chapter 7 itself.

All right.

So, anyone, just to begin with,

what were your thoughts or feelings on reading Chapter 7,

or what came to mind as you read or remembered reading Chapter 7,

Active Inference in Discrete Time?

Okay.

Just unmute and go for it.

Well, I didn't read it completely,

but there's a number of questions

that popped up in,
for instance, in Figure 7.3.

So, the A matrix is conditioned not only on one state,
but a couple of states.

Is that correct?

I mean, it's...

So, the sequence of preceding states
plays a role.

That's different to Hidden Markov Model, right?

And then...

So, this would be one question,
how this could be.

And then, another question is
whether this pie
actually makes this whole automaton more powerful,
or is this just a convenient way
of representing it?

So, could this pie actually be merged
into the state

as an additional factor or something?

So, multiply the number of states
and...

Or does it really increase
the computational power
of that whole automaton?

Great questions.

All right.

So, yeah.

Yeah.

Jakob, go for it first.

I think that the...

In terms of putting the pie
into the states themselves,

I don't think I've seen that ever
being done

in an actual model.

And I think this is more
of a representation

rather than really
how the model would be implemented
because the pie
is just...
is just a policy.
and it's...
I think it's...
It, like, branches out
to the two...
the two Bs
from, like, a single node
only to represent
that each B
is also conditioned
on the policy.
But not necessarily
that...
it's, like, a...
one...
that it happens
within, like,
one time snapshot.
Yes.
Yeah.
In...
in some other work,
Jakob and I have
talked a lot about, like,
why is pie
shown to intervene twice?
And even more generally,
why is it shown
with three time steps?
Yes, it helps us understand
that it's, like,
past, present, and future.
But the action perception loop
also could be shown

with just the transition
between two steps
and π intervening once.
Whereas this is
a little bit ambiguous
whether it's a policy
of length two
intervening at this time point
or whether it's just
showing the continual unfolding
of policy
interleaving
between the hidden states
in a sort of
open-ended way.

But...

what...

what do you mean,

Michael,

by increasing

the computational power?

I mean,

it...

it would be possible

to have

the...

dynamics

of the hidden states

through B

be complex,

but by removing

π

and the action selection

apparatus,

then

if we only had

the rake

down below,

it would just be
a passive inference
algorithm
with a potentially
complex
endogenous
dynamic.
well,
I...
I had to...
can I give an example?
Yeah.
Yeah.
So I...
I was thinking of...
so there is that
some kind of recursion
possible.
For instance,
I was thinking of
the following example.
Assume that I want to
go and buy bread
and then I go out
and then on my way
to the bakery,
I see a
haircutter
and I think,
oh,
before buying this bread,
I could go to the
haircutter
and then crossing the
street,
I see
a shop with
journals

and then I think,
oh,
when I sit there
with a haircutter,
I should maybe
buy a journal
to read something.
Then I go to that
journal,
I see a friend
and the friend
tells me,
oh,
I have a new car,
you want to come
to see my car?
And so I pass
the half of the day
going forward
and then eventually
I backtrack
coming back
from the car,
going to the
haircutter,
buying the bread
and in the evening
I go home.
So this is a kind
of a recursive thing,
right?
So there is an action
and then there is
an intervention
with another action
but then there is,
so it's a kind
of a recursive thing

and each of those
actions maybe
follows a policy
but that can be,
there can be another
policy that intervenes
but then I backtrack
from there.

So that would actually
be, yes,
a recursive kind
of a policy.

Is something
like this possible?

So this would really,
I guess,
increase the
computational power
with respect
to just,
completing one policy
or one sequence
of actions
and then going
to the next
where, you know,
where one iterates
the stuff instead
of the possibility
to do that recursively.

So something like this
I was thinking.

So is it possible
to have some recursive
policies there?

That's what I mean
with competition power.

One quick note

is to livestream 42
which was on robotics
and the SLAM
simultaneous localization
and mapping.

There's a higher
level policy selection
around what location
the robot wants to go
and then a lower level
tactical level
that is about
the actual
motor actuation.
location.

So in that example
that you gave
there might be like
a higher level model
of which location
you want to go towards
and then there's
a lower level
footstep model
and then
this
shifting of decision
could be modeled
as policies
on
which location
one is seeking.

so as with
many features
it's not necessarily
intrinsic
to this

figure 7.3

as the kernel
but you can compose it
in order to capture
some of those dynamics.

Ali

and then Jakob.

Yeah, well

actually

I think

described

in those terms

I mean

recursively

updating

the generative

model

is just

for the

it can just be

for the purpose

of building

a comprehensive

generative model

I mean

as comprehensively

as we can

but

when we include

the policies

as the additional

parameter

in our generative

model

actually

this is

a kind of

a technique

used

for
bringing
the elements
of action
selection
to
or
within
the
Bayesian
belief
updating
which is called
the planning
as inference
so
in order
to do that
policy
selection
or
I mean
constructing
space
of
all
possible
actions
is a necessity
a requirement
to do
these
kinds of
planning
as inference
but yeah
it's just
a kind of

mathematical
trick
if you will
to
somehow
bring together
the action
selection
into
the mechanism
of
Bayesian
belief
updating
change
but
but this
this
possibility
oh sorry
continue
yeah so
so this
possibility
is not
in picture
7.3
right so
because these
arrows go
from π
to the
B matrix
but there's
not a kind
of an arrow
going back
so

so
if
are you
saying
the trick
would be
that
one learns
a pi
inside
a pi
somehow
yeah
you see
this is just
one policy
a picture
a snapshot
of a one
policy
but
in
more
comprehensive
models
we have
the state
of
I mean
we have
policies
and not
just one
single
policy
to select
from
so

yeah
that's
just
that picture
just shows
how
the
I mean
choosing
a single
policy
can affect
the
generative
model
and
how
it can
do
the inference
via
planning
or action
selection
but
in reality
or
I mean
when dealing
with more
comprehensive
models
we have
not
we don't
necessarily
have just
one single

policy
to do
to
select
from
I mean
it would
be
tautological
to have
just one
single
policy
so we
need
a space
of
possibility
if
there
was
just
one
policy
and
there
was
no
agency
around
selecting
it
then
it
reverts
to a
passive
inference

challenge
whereas
the structure
of the B
matrix
which
you can
explore
from a
hands-on
perspective
like by
looking at
the
dimensionality
of it
in
PyMDP
the
inference
problem
is set
up
such
that
agents
are
deciding
between
 π_1
and π_2
which are
evaluated
according to
their expected
free energy
and π_1
and π_2

do different
things
to B
so we can
think of
like a
B1
and B2
not
time point
1 and 2
but
variant
or slice
1 and 2
that
change
how hidden
states
change
their time
so
Jakob
I guess
on the
question
of
like
recursion
and
with your
example
of like
going
to
a barber
and then
seeing

oh
I want
to buy
a magazine
to read
and
like
going
the policy
changing
throughout
I guess
I'm not
I'm not
really sure
I see
the recursion
there
in
like an
active
inference
model
because I
think it's
more
I'd say
it's more
like a
very high
dimensional
model
where you
are perhaps
performing
where you're
always like
narrowing down

your policy
search space
and with
new observations
you're like
adjusting
that search
space
based on
your preferences
in that
particular
observational
modality
like you
get reminded
that there
are magazines
and you can
buy magazines
to read
while you're
at another
place
so it
kind of
extends
that computation
but I don't
think it's
necessarily
recursive
and also
I think
the figure
I think
you mentioned
something about

like
prediction
errors
or
something
like that
I don't
think that's
necessarily
encapsulated
in the figure
because that
only shows
well
I guess
it's not
explicit
but we can
assume that
the B
there is
static
and
you can
imagine
a lot
of other
things
happening
if you
would be
learning
the B
matrix
or doing
model
selection
as well

and adjusting
your policy
based on
what
observational
modality
has the
highest
precision
Can I
answer
to this?
So I
think
well maybe
you can
represent
this
in a
way
of
iteration
or
the way
you say
but
it would
be
recursive
in my
understanding
of
these
terms
if
going for
instance
to the

baker
is a
policy
means
it's
a
series
of
actions
right
and so
if
this
is a
policy
and
then
in
the
middle
of
this
there
is
somehow
another
policy
that
pops
up
I
go
first
to
the
barber
or
to

the
barber
whatever
and
then
this
one
policy
still
exists
but
inside
there's
another
kind
of
sequence
of
actions
another
plan
and so
on
popping
up
and
this
could
happen
recursive
I
mean
in
my
understanding
of
this
term

of
these
terms
it
would
be
kind
of
recursive
yeah
it
can
be
through
nested
modeling
which
is
one
canonical
way
that
recursion
is
implemented
so
it's
unfolding
sequentially
through
time
but
also
like
this
higher
level
discrete

time
so
again
we're
chapter
seven
we're
in
discrete
time
and
so
this
might
be
a
slower
click
so
maybe
the
footsteps
are
taking
place
on
a
second
to
second
policy
tick
timeline
whereas
a
policy
in which
location

one wants
to go
is being
evaluated
at a
slower
level
and
if
this
nested
level
is
playing
out
at
a
similar
frequency
to
the
lower
level
then
as a
modeler
you might
be
better
off
just
having
it
as
a
single
level
model

and
as
an
organism
you'd
get
into
these
kind
of
like
paralysis
cases
where
if
you
were
actually
switching
policies
about
where
you
want
to
go
then
this
would
be
incoherent
down
in
the
lower
level
so
just

return
to
the
initial
question
is
it
possible
to
merge
policy
into
the
hidden
states
as
a
causal
latent
factor
well
in
a
way
policy
intervenes
in
how
hidden
states
change
through
time
so
policy
does
influence
hidden

states
without
an
action
selection
apparatus
it's
not
active
inference
it's
not
even
a
control
theory
problem
it's
just
a
question
of
identifying
latent
causes
from
the
unfolding
of
hidden
states
and
observables
that
you
can't
control
and

that's
a
classical
SPM
type
challenge
like
S
might
be
the
latent
unobserved
neural
activity
and
O
might
be
multi-sensor
fusion
with
fMRI
EEG
and
MEG
so
this
can
still
be
very
complex
in
its
unfolding
through
time

with
the
observations
and
the
hidden
states
what
would
make
it
an
active
inference
problem
well
if
you
weren't
just
observing
sensor
modalities
but
if
the
experimenter
could
take
a
policy
that
was
intervening
how
neural
dynamics
change

through
time
which
is
what
experimental
design
is
which
is
why
it's
so
compatible
with
thinking
about
the
meta-Bayesian
approach
with a
behavioral
researcher
so
pi
influences
hidden
states
but
if
we
merely
folded
into
B
we
would
lose

the
ability
to
do
actual
action
selection
but
if
it
were
a
passive
inference
problem
we
still
could
use
variational
inference
we
still
could
use
free
energy
we
still
could
even
align
outcomes
with
our
preferences
we
could

still
calculate
and
bound
surprise
the
active
embedding
of
active
inference
is
to
include
action
as
a
parameter
like
Ali
mentioned
within
the
Bayesian
brain
context
to
have
a
parameter
that
you
do
Bayesian
inference
on
and
then

extend
that
not
just
from
perception
and
causal
inference
which
is
what's
happening
on
the
bottom
like
perception
is
the
a
causal
inference
can be
understood
as
B
but
action
selection
does
require
an
additional
apparatus
passive
inference
more

like
signal
processing
action
more
like
control
theory
unified
imperative
so
the
G
would
then
be
predicting
the
future
expectations
developing
of
expectations
and
so on
what
would
it
be
what
is
G
the
G
the
pie
sorry
well

pie
is
the
combinatorics
of
affordances
over a
given
time
horizon
so
merely
generating
pie
as a
list
is not
intrinsically
difficult
it's just
the search
space
of
possible
action
sequences
over a
given
time
horizon
if
there's
two
options
and
you're
doing
a

time
horizon
of
three
then
it's
two
cubed
G
is
the
expected
free
energy
which
updates
the
policy
prior
so
take
that
list
and
it's
your
prior
probability
of
those
policies
which
could be
either
just
a
uniform
prior

or
could
reflect
habit
like
a
priori
how
likely
our
policies
and
then
G
sharpens
or
updates
that
policy
vector
into
a
policy
posterior
which
then
the
agent
samples
their
actions
from
and
and
how
how is
this
meant

so
assume
that
I'm
so
you're
saying
the
first
is
the
past
and
then
so
is
this
re
estimated
at
every
state
so
so
this
pi
would
somehow
have
expectations
over
future
affordances
as you
say
right
would
say

which
transitions
are more
likely
in the
to the
right
of this
graph
and
would
that
be
re-estimated
in
at
each
time
time
step
or
how
would
how
what
that
yeah
there's
there's
multiple
possible
levels
Ali
go
first
then
I'll
give

a
thought
well
basically
pi
actually
is just
an
additional
parameter
multiplied
to
the
basic
Bayes
theorem
formula
we had
before
so
instead
of just
multiplying
likelihood
and
the
priors
simply
we
also
multiply
the
probability
of
selecting
a
policy
and

that's
just
it
so
pi
is
just
an
additional
parameter
in that
equation
from
chapter
back
in chapter
two
yeah
equation
2.6
is where
g is
defined
so
g
is a
function
of
policies
so
it
takes
in
as
arguments
the
vector
of

policies
and
then
conditioning
as we
see
in
every
single
of
these
rephrasing
every
single
one
is
conditioned
upon
a
given
policy
and
then
different
policies
are
compared
by
their
expected
free
energy
and
then
depending
on
the
precision

on
the
action
posterior
which
isn't
shown
in
7.3
you
can
have
a
spectrum
of
behavior
if
you
have
extremely
high
precision
like
it
could
be
thought
of
as
low
temperature
in
that
case
one
policy
that's
let's

just
say
one
is
it's
1%
better
than
the
other
then
under
the
low
temperature
asymptote
the
high
precision
asymptote
the
policy
that's
even
tiny
tiny
bit
better
will
always
be
selected
under
the
neutral
temperature
zone
policies

are
selected
based
upon
their
posterior
likelihood
so
if
one
of
them
is
51%
and
the
other
is
49%
when
you
soft
max
it
then
it's
selected
51%
and 49%
of the
time
respectively
and
then
in
the
high
temperature

or
low
precision
asymptote
also
known as
shaky
hand
the
differences
amongst
policies
in their
expected
free
energy
basically
get
erased
and
at
that
point
policies
are
selected
essentially
without
respect
to
their
posterior
likelihood
it
this
this
is
one

of
the
key
pieces
though
because
again
we're
not
calculating
different
policies
and
asking
how
rewarding
are
they
how
much
expected
reward
do
they
have
and
then
ranking
or
selecting
based
on
that
rather
there's
a
unified
imperative

expected
free
energy
which
under
different
decompositions
can be
shown
to have
these
different
interpretations
that is
then
used
to
generate
a
policy
posterior
and
then
as
just
described
that
policy
posterior
in
the
context
of
a
temperature
or
precision
variable

can
exhibit
behaviors
ranging
from
always
selecting
the
slightly
preferred
policy
in
the
low
temperature
limit
neutral
temperature
selecting
according
to
the
posterior
high
temperature
erasing
differences
in
the
posterior
yeah
sorry
I just
wanted to
point out
that
this
G

matrix
here
is
still
prior
to
learning
so
if
we
want
to
add
learning
to
the
policies
things
would
get
a little
bit
more
not to
say
complicated
but
more
extended
equations
would
be
needed
so
we
haven't
included
learning

in this
specific
formulation
yet
it's
prior to
learning
yes
thanks
yeah
I
also
want to
highlight
that
like
in
the
category
of
phenomena
that we
can
model
but
are
not
just
simply
in
this
image
you
mentioned
like
learning
new
policies

so
it's
like
in
the
fixed
fixed
fixed
case
you're
still
able
to
do
action
selection
but
you're
not
learning
or
changing
which
policies
you
can
enact
you're
not
changing
what
affordances
you
believe
you
have
nor
are

you
updating
B
which
is
to
say
you're
not
updating
your
beliefs
about
how
policies
influence
hidden
state
changes
the
next
level
of
introduction
of
novelty
is
learning
on
B
where
you
of
sentences
and
to
introduce
recursive

referential
clauses
or
learning
it's
a more
complex
sentence
so to
speak
but
this
is
the
archetype
that
sets
the
stage
for
what
cognitive
model
structures
could
exhibit
that
kind
of
phenomena
Michael
yeah
so
a
policy
would
be
a

number
of
actions
right
would
be
a
sequence
of
states
is
that
is
my
understanding
correct
and
if
this
is
the
case
then
a
policy
would
select
a
number
of
states
and
my
question
I guess
is
so
then

I
come
at
the
end
or
at
what
point
are
these
policies
then
calculated

so
I
come
to
the
end
of
an
action
and
then
only
is
the
new
policy
a
computer
or
selected
or
is
it
constantly

the
case
so
if
I
decide
to
buy
a
bread
for
instance
and
then
I
go
out
and
I
cross
the
street
and
do
this
and
that
so
there
is
a
couple
of
actions
involved
right
and
so

my
decision
to
buy
a
bread
was
done
before
but
on
the
way
is
that
policy
recalculated
and
maybe
on
the
way
I
think
something
else
is
more
reasonable
to
do
okay
so
again
that's
a
slightly
nested

scenario
with
multiple
levels
of
policy
that
you're
looking
at
and
then
just
to
be
clear
so
policy
it's
a
sequence
of
actions
so
let's
just
say
that
the
affordances
there's
two
affordances
up
or
down
and
then

time
horizon
is
three
well
let's
just
say
it's
two
there
are
four
policies
UU
UD
D U
D D
so
generating
the state
space
of
policies
is
trivial
it's
just
all
of
the
strings
of
affordances
now
you
ask
where

does
policy
inference
happen
where
and how
does it
happen
and there's
a few
different
answers
so
first
off
here's
from
our
active
inference
and
modeling
conflict
paper
here
we chose
to show
only a
two-step
model
so
instead
of
the
three
because
we
were

anticipating

that

ambiguity

where

pi is

intervening

twice

when

actually

control

is only

exerted

at any

given

moment

but

it's

anticipated

and

then

we map

this

onto

the

UDA

observe

orient

decide

act

this

isn't

a

formal

mapping

per

se

but

it

helps
understand
sequentially
for
example
in a
reactive
message
passing
paradigm
what
different
informally
categorized
states
of the
perception
cognition
action
cycle
how that
maps
onto
this
Bayesian
graph
so
since
states
come
in
cognition
occurs
or
one could
also
take a
broader

perspective
then
but
this
updates
one's
beliefs
about
the
world
then
there's
an
inference
on
action
so
here
are
the
antecedents
of
policy
selection
which
are
E
what can
be done
the
affordances
C
what is
preferred
the
preferences
preferred
slash

expected
and
then
with
C
and
E
as it
shows
here
G
is a
function
of
policies
conditioned
upon
C
and
E
so
with
those
antecedents
taken care
of
we can
update
our
beliefs
on
pi
to
move
from
a
policy
prior

to
a
policy
posterior
those
policy
posteriors
can
then be
sampled
from
as
discussed
earlier
at
high
low
or
neutral
temperature
and
then
the
selected
policy
intervenes
and
ends up
choosing
a
slice
in
B
and
then
the
slice
of

B
that's
chosen
influences
how
the
hidden
states
change
for
example
if
there's
two
policies
on
a
one
time
set
basis
so
same
as
the
affordances
in
this
case
turn
on
the
heater
or
do
not
turn
on

the
heater
so
maybe
if
we
don't
turn
on
the
heater
then
the
B
matrix
just
passes
the
hidden
state
forward
the
temperature
doesn't
change
if
we
do
turn
on
the
heater
the
temperature
increases
immediately
again
simple

example
so
we
looked
at
the
thermometer
O
updated
our
belief
given
our
priors
about
the
temperature
of
the
room
then
with
respect
to
what
we
can
do
E
and
what
we
expect
slash
prefer
C
B
we

updated
our
policy
prior
into
a
policy
posterior
we
then
selected
the
action
that's
most
likely
not
the
most
rewarding
action
the
most
likely
action
and
that
influenced
how the
hidden
states
change
through
time
and
then
the
UTA

cycle
begins
again
action
intervenes
in the
discrete
time
formalization
which is
where we're
in chapter
seven
it
intervenes
in every
time
click
okay
thank you
so much
and
and also
um
the
pragmatic
turn
is um
it's a book
co-edited
by
Friston
but it
discusses
like
what
has
happened

in
neuroscience
with
people
embedding
!
action
and
thinking
about
action
in
neuroscience
and
that's
relatively
recent
which is
really
exciting
okay
but these
are
these
are
great
questions
and
then
let's
just
also
go
to
the
second
question
that

you
raised
there
and
then
we'll
scan
through
all
of
chapter
seven
so
is
the
A
matrix
in
figure
7.3
conditioned
on
one
state
or
a
series
of
states
so
yes
Jakob
feel free
to add
any more
details
on this
one

but
as
it's
shown
here
from a
Bayes
graph
perspective
it
is
O
of
t
minus
one
is
only
conditioned
on
s
of
t
minus
one
but
when
we
think
about
across
the
vector
of
observations
they
can be
understood

as being
conditioned
upon
the
time
series
of
hidden
states
so

So
each
observation
is only
conditioned
upon
its
associated
hidden
state
that's
the
partially
observable
and
hidden
states
pass
through
the
Markov
needle
of
the
present
that's
the

Markovian
property
past
only
influences
the
future
through
the
present
which
isn't
to say
that's
how
reality
is
that
just
to
say
it's
a
simplifying
structural
assumption
that
one
could
vary
if
they
chose
but
it
is
a little
ambiguous

I
can
see
why
somebody
might
think
that
observation
at a
given
t
let's
say
t
plus
one
is
a
function
that's
conditioned
upon
all the
previous
times
but
not
directly
observations
are like
cul-de-sacs
and
again
a
is
the
tail

of
two
densities
the
recognition
density
which
is
from
observations
updating
beliefs
about
hidden
states
bayesian
recognition
model
and
the
generative
density
which
is
from
a
hidden
state
generating
pseudo
observations
and
the
way
that
those
two
densities

can be
alternated
is
used
and
has
been
used
for
decades
in
expectation
maximization
algorithms
okay
cool
great
questions
here
um
does
anyone
want to
go to
any
of
the
chapter
seven
questions
or
should
we
scan
through
in
the
last

15
minutes
here
just
like
what
happens
in
chapter
seven
all
right
let's
scan
raise
your
hands
when
you
want
and
I'll
immediately
go to
you
if
you
want
to
um
just
add
a
note
on
like
as
we

scan
through
so
we
will
scan
through
the
whole
thing
in
the
next
ten
minutes
okay
starting
with
the
quote
what
I
cannot
create
I
do
not
understand
famous
quote
take
it
or
leave
it
but
now
we're

into
the
building
the
recipe
has
been
disclosed
in
chapter
six
the
prerequisite
knowledge
has been
distributed
in chapters
one
through
five
ideally
they
start
with
perceptual
processing
this
is
very
much
like
the
step
by
step
paper
step
by

step
starts
even
simpler
with
starting
with
just
a
static
inference
but
here
they
go
immediately
to
dynamic
perceptual
inference
this
is
just
the
bottom
of
the
rake
this
is
a
passive
inference
but
it
is
changing
through

time
they're
going
to be
using
a
music
example
speaking
to their
target
audience
which
is
Ali
a
musical
example
so
we're
listening
to
some
music
not
necessarily
playing
the
music
but
the
outcome
is the
note
that's
actually
intended
I'm

sorry
the
outcome
is what
is
actually
played
and
the
hidden
state
is
the
intended
note
and
just
from
looking
at
this
A
matrix
we
can
see
that
like
on
the
diagonals
if
it
was
the
identity
matrix
that

would
mean
that
there
was
a
perfect
mapping
between
hidden
state
and
observations
but
this
is
not
an
identity
matrix
it
has
off
diagonal
values
which
is
to
say
it's
like
a
blurry
A
matrix
and
in
the

limit
of
a
blurry
A
an
ambiguous
A
you
can
imagine
if
it
was
all
2.5
across
then
there
would
be
no
information
on
hidden
state
given
the
outcome
the
other
extreme
would
be
perfect
information
with
the

identity
matrix
and
you
can
see
that
it's
also
rescaled
so
the
numbers
can
be
shown
simply
as
integers
and
that
it's
rescaled
so
it's
a
proper
probability
70%
of the
time
the
musician
hits
their
intended
note
row

1

column

1

7

B

is

the

probability

of

transitions

so

we

can

go

we

go

from

note

1

to

note

2

with

97

so

notice

the

diagonal

if

that

was

the

identity

matrix

on

B

if

that

was
all
B
was
you
would
basically
dwell
wherever
you
began
if
B
is
all
off
diagonal
you
never
stay
in the
same
spot
twice
and
so
here
is
a
sequence
of
notes
because
1
almost
always
moves
to

2

2

almost

always

moves

to

3

3

almost

always

moves

to

4

and

4

almost

always

moves

to

1

then

D

is

the

prior

and

so

it's

just

being

taken

as

a

noiseless

estimate

that

the

first note

is the
beginning
note
Michael
so

is
this
a
very
boring
tune
that
plays
four
notes
in the
in the
round
eternally
exactly
like this
it's like
G-E-B
in eternal
golden
braid
except
it's
like
G-E-B
X
G-E-B
X
okay
very boring

yeah
okay
thank you
yes
it is a
simple
one
so
7.1
to 7.3
equations
completely
specify
the
HMM
note
that
it's
a
hidden
Markov
model
because
it has
this
partially
observable
character
but
it's
not
a
decision
process
because
there's
no
action

selection

but

all

we

needed

to

do

was

to

specify

A

B

and

D

so

to

totally

specify

the

HMM

was

ABD

we

didn't

have

to

say

which

notes

we

preferred

we

didn't

have

to

say

anything

about

what
we
could
do
with
C
or
E
okay
then
there
are
some
simulations
shown
there's
a few
things
about
this
simulation
first
as
an
earlier
question
has
raised
why
are
there
continuous
interpolations
when
we're
in the
discrete
time

world
a
general
comment
is
these
graphs
with
multiple
block
lines
are
a
little
hard
to
interpret
because
of
course
we
can't
tell
which
line
is
which
another
issue
with
7.2
figures
specifically
is
the
bottom
left
quadrant

is
empty
so
just
a few
points
but
what's
basically
happening
is
in
this
simulation
run
which
would
only
happen
this
way
several
percent
of the
time
but
they're
showing
it
basically
in
the
upper
right
the
retrospective
beliefs
are

shown
so
the
retrospective
beliefs
are
shown
with
a
correct
song
even
though
the
actual
observations
were
note
one
two
two
four
so
this
observation
was
recognized
and it
did
increase
the
posterior
on
note
three
being
played
however

because
the
belief
was
so
strong
that
note
three
would
be
played
in
this
time
it
did
not
move
the
needle
far
enough
to
actually
hear
that
note
so
that's
like
when
somebody
hears
something
that
isn't
said

because

it's

likely

that

it

was

said

okay

so

that's

a fun

model

but

it's

not

action

inference

yet

so

that's

where

we get

to

7.3

where

we've

been

spending

a lot

of

today's

discussion

7.3

discusses

the addition

of an

extra

variable

on which
transition
probabilities
are
conditioned
so
it's
like
really
when
we
think
b
we
should
really
think
b
vertical
line
pi
because
the
inference
on
action
means
that
we
have
actual
options
for
action
which
is
to
say

there
are
multiple
b's
that
can
influence
how
s
unfolds
so
we're
really
talking
about
b's
condition
on
policy
unless
we're
talking
about
the whole
b
caterpillar
!
the whole
tensor
of b
b's
okay
here's
where
expected
free
energy
and

one
of
the
partition
is
shown
which
is
discussed
in
chapter
2
equation
2.6
and
in
chapter
4
so
next
week
when
we're
in
chapter
7
for
the
second
time
let's
return
talk
a little
bit
more
about
the

technical
parts
of
what
exactly
equation
7.4
means
it's
a
decomposition
that
enables
epistemic
drive
towards
information
gathering
that's
this
first
part
of
the
right
hand
side
and
the
pragmatic
drive
which
is
about
realizing
prior
beliefs
reducing

surprise
about
prior
beliefs
not
maximizing
a reward
function
reducing
divergence
between
preferences
and
outcomes
they're
going
to
unpack
the
full
7.3
figure
in
the
context
of
a
T-maze
which
is
a
rat
that's
able
to
go
left
or

right
to
find
food
or
not
or
it's
able
to
get
a
queue
that's
the
queue
location
R
there's
four
locations
that
it
can
be
in
and
there's
modeler
degrees
of
freedom
and
how
different
things
are
represented

so
this
isn't
even
the
only
way
to
represent
this
toy
example
but
this
matrix
does a
really
good
job
of
showing
what
are
the
states
what
are
the
dimensionality
of
the
difference
variables
which
we
can
come
to

next
week
so
here's
context
two
here's
context
one
here
we
have
black
on
the
right
side
white
on
the
left
here
we
have
black
on
the
left
white
on
the
right
and
there's
some
subtle
differences
which

you can
note
like
specifically
with
lines
two
and
three
here's
a
one
here
here's
a
one
on
the
third
line
because
the
queue
is
suggesting
that
it's
on
the
right
here
the
queue
is
showing
it's
on
the

left
and
in
model
stream
7.2
that
example
is
like
very
hands-on
and
in
some
ways
even
better
than
this
one
and
it
includes
things
like
learning
but
it's
a
very
similar
task
here
is
B2
so
they're

showing
the
dimensionality
of
what
those
B
slices
look
like
for
different
policies
this
is
representing
the
B
tensor
here
are
preferences
this
is
what
a
preference
vector
looks
like
and
in
step
by
step
the
paper
and

model
stream
1
series
and
in
model
stream
7
series
there's
some
discussion
on
what
does
C
mean
is
it
being
reflected
as
log
probabilities
why
would
one
do
6
and
negative
6
versus
6,000
and
negative
6,000

or
0.6
and
negative
0.6
what
do
those
values
actually
mean
and
the
priors
provided
so
now
that
we've
provided
a
controllable
B
which
is
the
manifestation
of
policy
options
and
provided
a
C
we
can
do
active

inference
because
we
have
now
what
we
need
those
two
pieces
really
that's
what
brought
us
from
the
HMM
to
the
POMDP
was
what
we
can
do
and
what
we
prefer
section
7.4
goes
into
information
seeking
it

unpacks
the
epistemic

value
into
a few
more
sub
decompositions
and
it
talks
about
how
that
plays
out
there's
some
nuance
here
again
there's
also
some
semi
confusing
black
and
white
graphs
but
other
ones
are
clear
we

can
talk
about
how
epistemic
value
is
being
realized
in
this
example
box
one
is
a
bit
of
a
side
note
on
uncertainty
and
precision
and
here's
showing
really
the
temperature
or
the
precision
highest
temperature
lowest
entropy

most
uniform
distribution
lowest
temperature
highest
entropy
most
specific
distribution
here
you're
sharpening
here
you're
blurring
okay
7.9
as mentioned
two pages
earlier
is an
eye
cascade
so
there are
four
locations
and the
eyes
are
circading
again
one could
ask about
where are
these
continuous

interpolations
coming from
since we're
in chapter
7
in a
fully
discrete
situation
but
it's
all
good
so
eye
circadian
different
sequences
or
traces
of
eye
circades
across
four
locations
driven
by
epistemic
value
in
section
7.5
learning
and novelty
is
introduced
so

as
we
were
mentioning
earlier
learning
with respect
to the
A or
the B
matrix
so far
we've been
talking about
them as
fixed
here
is where
learning
happens
on
A and
B
and just
like you
would do
with any
Bayesian
inference
whether on
action
or on
the consequences
of action
or on
perception
learning
is modeled

as a
hyperprior
or a
variable
that
governs
a prior
on another
parameter
so here
is like
a parameter
that
is adding
a layer
to
the rake
we still
see the
core
figure
4.3
figure
7.3
motif
but there
are these
new
little
edges
that are
the
learning
edges
so it's
the same
POMDP
as figure

7.3

but the

priors

for each

hidden

states

now depend

on the

variables

which

themselves

come with

prior

beliefs

so the

prior

on

the

variables

a and

b

is

in

the

Bayesian

paradigm

how we

learn

or update

our values

on a

and b

brief discussion

on conjugate

priors

some more

equations

that we

can come
to next
week
about
updating
and learning
on a
another
look
in
equation
7.11
on
g
which
is
very
similar
to
what
we
saw
in
equation
2.6
different
partitionings
on
expected
free
energy
policy
then
there's
a
final
example
that's

now
in
a
maze
and so
this
is
some
maze
seeking
entity
we can
come to
this
next
week
in
hierarchical
inference
hierarchical
or deep
inference
something we
kind of
mentioned
today
but we'll
look more
at next
week
is
first
there's
a box
on
structure
learning
which

as
mentioned
earlier
is when
we
think
about
regular
non-structure
learning
as like
fine-tuning
quantitative
parameters
within
a given
dimensionally
defined
generative
model
structure
learning
and what
you can
do
with
structure
learning
like
specifically
Bayesian
model
reduction
is when
we're
actually
modifying
the

dimensionality

of the

generative

model

that's

in

box

7.3

here

is

a

nested

generative

model

which is

almost

exactly

the

form

that we

looked

at

from

live

stream

42

where

we

see

like

the

rake

classical

figure

4.3

figure

7.3

paradigm

and then
there's
like a
meta
rake
where
here's
another
E
downward
facing
and we
have
policy
at a
higher
level
and
different
generative
models
have
taken
different
interpretations
on what
the lower
and the
higher
level
are
like
in
live
stream
42
the
lower

level
was
tactical
movement
of a
robot
and the
higher
level
was
strategic
location
preference
whereas
in
live
stream
28
!
another
example
with
decision
making
we'll
come to
and then
the chapter
summarizes
so
thank you
for
this
session
we'll
stop
recording
and take

a break

before

the next

one

will

be

be